

Airbnb Dynamic Pricing Recommendation Engine

PROJECT REPORT

Project: Airbnb Dynamic Pricing Recommendation Engine

Tools Used: Python, Excel, Power BI

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INTRODUCTION

- This project focuses on building a dynamic pricing recommendation engine for Airbnb listings using historical data.
- By analyzing location, seasonality, and listing quality, it helps suggest optimal prices to maximize revenue and occupancy.

ABSTRACT

- This project builds a dynamic pricing recommendation engine for Airbnb using Python and Power BI.
- It analyzes factors like location, room type, and reviews to suggest optimal prices for maximizing revenue.

WORKFLOW

- **Data Collection:**
 - Dataset: 'listings.csv' with Airbnb data (location, price, reviews, etc.).
- **Data Cleaning in Python:**
 - Removed rows with missing price.
 - Selected features: price, neighbourhood_group, room_type, number_of_reviews, reviews_per_month, availability_365.
 - Handled missing values in reviews_per_month.
 - Converted categorical columns to dummy variables.
 - Exported cleaned dataset to 'airbnb_cleaned.csv'.
- **Data Analysis in Excel:**
 - Checked columns, formatted as table.
 - Saved final version as CSV (UTF-8 format).
- **Power BI Dashboard:**
 - Imported 'airbnb_cleaned.csv'.
 - Used Transform Data to split by comma and used first row as header.
 - Built visualizations: price trends, distribution by location, room types.

RECOMMENDATIONS

➤ **Optimal Pricing Insights:**

- Listings in Manhattan and Brooklyn can be priced higher due to demand.
- Entire home/apt types earn more than shared/private rooms.
- Listings with higher reviews_per_month tend to have better occupancy.

➤ **Suggestions:**

- Adjust pricing based on season and area (use slicers in Power BI).
- Improve room quality and collect reviews to justify higher pricing.
- Use Power BI dashboard monthly to re-evaluate pricing based on data trends.